Cloud Data Centers: Finding the Balance Between Energy Savings and System Speed

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As businesses rapidly migrate to the cloud and global Artificial Intelligence (AI) infrastructure expands, the energy demands of cloud data centers are reaching record highs. This study aims to evaluate tradeoffs in power, efficiency, and cost in cloud infrastructures. Utilizing the CloudSim Plus framework, this research models data center environments from different time periods to determine if more modern systems reduce electrical costs without sacrificing speed and efficiency. The methodology includes a multi-host simulation, to represent a larger scale operation. This paper calculates and compares the total energy consumption, efficiency, and estimated operational costs of each data center.

CCS CONCEPTS • Computer systems organization~Architectures~Distributed architectures~Cloud computing • Hardware~Power and energy~Power estimation and optimization

Additional Keywords and Phrases: Cloud computing, Energy efficiency, CloudSim Plus, Data center cost analysis, Power Usage Effectiveness.

1 INTRODUCTION

Artificial Intelligence (AI) cloud computing infrastructures are known to consume massive amounts of energy to operate. As national infrastructure continues to build at a rapid pace, the power required to maintain these servers is hitting unprecedented levels. Reducing this energy footprint is important not only for environmental sustainability, but also for the economic viability of organizations relying on these services.

The primary objective of this research is to evaluate fundamental cloud infrastructure concepts to better understand the tradeoffs in power, efficiency, and cost. Specifically, this study investigates whether it is possible to reduce the electrical costs of cloud data centers without compromising the speed and efficiency necessary for end-users. To achieve this, a multi-host simulation was designed and executed using CloudSim Plus, a popular framework for modeling cloud computing infrastructures and examining their power consumption.

2 BACKGROUND AND RELATED WORK

While server power requirements saw an increase from 2000 to 2005, demand remained relatively constant in the decade following 2005 [8]. However, the current “race to build national AI infrastructure [3]” has drastically shifted this standard. The International Energy Agency projects that by 2030, data centers will account for 3% of total global electricity consumption, largely driven by the demands of AI [4]. As a result, experts warn that the United States' current energy infrastructure faces many challenges with the demand for new data centers [13].

To address these sustainability challenges without risking physical hardware, researchers frequently rely on simulation tools. For example, one study utilized CloudSim Plus to test a “Green Cloud” framework [5]. Their simulations demonstrated that employing virtualization for carbon-aware workload migration can significantly lower a data center's carbon footprint while maintaining high operational efficiency.

CloudSim Plus is widely used in cloud computing research as it is a popular fork of CloudSim, the first open-source specialized cloud simulation framework [7]. CloudSim Plus improves its predecessor in several ways, but most importantly the objective of this paper is its Power package, which contains classes that allow for power-aware simulations, and power consumption models [10]. The framework also gives users the ability to specify parameters about the hardware such as maximum active power, memory, and processing speeds, which is crucial for this study's comparison of different data center types.

To start creating the desired simulation, a `Datacenter` and `DatacenterBroker` are initialized. The `DatacenterBroker` can manage all `Datacenter` objects, so, for this study, each `Datacenter` has a dedicated `DatacenterBroker` to separate the three data center types. Along with the creation of `Datacenter`, a `Host` list is also created. The Virtual Machine (`Vm`) list is created after and shares the same hardware specifications given to its corresponding `Host` (Legacy, Modern, or AI). The `Vm` list is submitted to the `DatacenterBroker`. Cloudlets are created next, and the user can set up the number of tasks as well as the task length. Once completed, the `Cloudlet` list is also submitted to the `DatacenterBroker`, which ultimately executes the simulation.

The main advantage of utilizing CloudSim Plus for this study is the customization options for the `Host` and `Vm` class objects. It is even possible to adjust the timing of the tasks with the `schedulers` class, and to detail how a given resource is used by an application during its execution with the `utilizationmodels` class. This study sticks to a more simplified simulation, but the possibilities this framework allows are vast.

3 METHODOLOGY

To test the varying data center without risk to physical hardware, the CloudSim Plus framework was employed via the Eclipse Integrated Development Environment (IDE). The simulation environment evaluates cloud computing efficiency based on power and energy metrics. To add more realism to the simulation, actual server specification benchmarks were researched and implemented in the program for each system as shown in Table 1.

Table 1: Data Center Server/Host Specifications

Benchmark Specification	Units
Random Access Memory (RAM)	Megabytes (MB)
Storage	MB
Bandwidth (BW)	Megabits per second (Mbps)
Processing Elements (PE) / Cores	Million Instructions Per Second (MIPS)
Active / Idle power draw at 100%	Watts (W)

3.1 Experimental Setup

The experimental design incorporates data centers with legacy, modern, and AI hosts. A simplified program allowed for testing the framework. After successfully running the simplified simulation, I modified it to increase the host and task numbers. The final multi-host simulation has 100 hosts (the maximum allowed before experiencing Java Runtime Error due to a lack of enough RAM for a bigger simulation), and 10,000 tasks with a length of 50,000 Millions of Instructions (MI). The data centers have hosts with specifications as shown in Table 2.

Table 2: SPECpower_ssj2008 Results

Legacy Enterprise Server **HP ProLiant DL360 G1** Benchmarked in 2000, [5]	RAM: 2,048 MB Storage: 36,000 MB BW: 10 Mbps PE (x2): 1,000 MIPS each Active power draw: 292 W Idle power draw: 204 W
Modern Enterprise Server **HPE ProLiant DL360 Gen10** Benchmarked in 2017, [11]	RAM: 98,304 MB Storage: 340,000 MB BW: 4,000 Mbps PE (x56): 2,500 MIPS each (two processors with 28 cores each) Active power draw: 451 W Idle power draw: 48 W
AI Enterprise Server **HPE ProLiant DL380a Gen11** Benchmarked in 2024, [12]	RAM: 262,144 MB Storage: 122,880,000 MB BW: 4,000 Mbps PE (x128): 15,000 MIPS each (two processors with 64 cores each) Active power draw: 736 W Idle power draw: 235 W

Barroso's 70% high-idle rule [1] was applied to find the idle power draw for the legacy server.

Only the HP ProLiant DL360 G1 had a recorded MIPS number in its benchmark. MIPS as a metric for measuring processor speeds is now obsolete, as it can change depending on factors like workload mix and memory cache sizes, among other factors. [6]. However, because the CloudSim Plus framework relies on it, the numbers for the modern and AI systems are estimated by dividing the system throughput at 100% target load from each benchmark by their total core count.

Simulation code snippet showing how the AI data center hosts are created:

```
private static Datacenter createMultiHostAI(CloudSimPlus simulation, int numHosts) {
    List<Host> hostList = new ArrayList<>();
    for (int h = 0; h < numHosts; h++) {
        List<Pe> peList = new ArrayList<>();
        for(int i = 0; i < 128; i++) {
            peList.add(new PeSimple(15000));
        }
        Host host = new HostSimple(3072000, 4000, 122880000, peList);
        host.setPowerModel(new PowerModelHostSimple(736, 235));
        hostList.add(host);
    }
    return new DatacenterSimple(simulation, hostList);
}
```

After executing the simulation, an output table shown in Figure 1 is generated with details relating to each task completed (cloudlet) such as the start and finish time per data center. For the multi-host simulation, as the program output totaled three million rows, a CSV exporter method was added to save the data instead. Since Microsoft Excel will not open files with more than one million rows, I chose to import the data into Power BI, shown in Figure 2, for evaluating energy usage and to calculate efficiency and costs. The total runtime for the program was 29 minutes and 52 seconds.

SIMULATION RESULTS

Cloudlet ID	Status	DC ID	Host ID	Host PEs CPU cores	VM ID	VM PEs CPU cores	CloudletLen MI	FinishedLen MI	CloudletPEs CPU cores	StartTime Seconds	FinishTime Seconds	ExecTime Seconds
1	SUCCESS	2	0	1	1	1	50000	50000	1	0.1	50.1	50.0
2	SUCCESS	3	0	1	2	1	50000	50000	1	0.2	20.2	20.0
3	SUCCESS	4	0	1	3	1	50000	50000	1	0.3	3.6	3.3

Figure 1: CloudSim Plus original output in Eclipse.

Cloudlet_ID	Data_Center	VM_ID	Status	Length_MI	Start_Time_sec	Finish_Time_sec	Execution_Time_sec
0	AI_Gen11_Cluster	3000	SUCCESS	50000	0.3	3.63	3
0	Modern_Gen10_Clust	2000	SUCCESS	50000	0.2	20.2	20
0	Legacy_2000_Cluster	1000	SUCCESS	50000	0.1	50.1	50
1	Legacy_2000_Cluster	1001	SUCCESS	50000	0.1	50.1	50
1	Modern_Gen10_Clust	2001	SUCCESS	50000	0.2	20.2	20
1	AI_Gen11_Cluster	3001	SUCCESS	50000	0.3	3.63	3
2	Modern_Gen10_Clust	2002	SUCCESS	50000	0.2	20.2	20
2	Legacy_2000_Cluster	1002	SUCCESS	50000	0.1	50.1	50
2	AI_Gen11_Cluster	3002	SUCCESS	50000	0.3	3.63	3

Figure 2: CloudSim Plus output imported into Microsoft Power BI.

3.2 Energy Evaluation

For the calculation of the total energy used by each data center type, the following formulas were applied [9]:

$$\exists_k = \{\Im \times [\Phi_i^R \times \eth_k^{active} + (1 - \Phi_i^R) \times \eth_k^{idle}], \Im \times \eth_k^{sleep}\} \qquad (1)$$

$$\exists = \sum_{k=0}^{n} \exists_k \qquad (2)$$

- $\Im$ = Execution Time
- Φ_i^R = CPU Utilization (a percentage from 0.0 to 1.0)
- $\eth_k^{active}$ = Active Power
- $\eth_k^{idle}$ = Idle Power

Equation 1 calculates the energy consumption of a single host *k*. Research shows that energy consumption has a linear relationship with processor utilization, with CPU consumption being the primary contributor to energy use. The formula assumes three server states: active, idle, and sleep.

Equation 2 defines the total energy consumption of a cloud data center, including all hosts. Due to the limited information available about the idle power of the legacy system, in this simulation the CPU was 100% fully utilized, which zeroes out the idle power and sleep portions of Equation 1. Thus, the total energy used per data center in this study is calculated as the execution time multiplied by the active power, 100 times for each host in the data center, shown in Equation 3:

$$\exists_{100} = \Im \times \left[1 \times ð_k^{active}\right] \qquad (3)$$

Table 3: Total energy used by Data Center calculation

Data Center	Execution Time	Active Power	Total Energy (J)
Legacy	50	292	1,460,000
Modern	20	451	902,000
AI	3	736	220,800

The findings in Table 3 match assumed expectations, as each data center had the same number of tasks with the same length, so although the AI data center uses more power, it executes the tasks in a very short period. The total energy used ended up being the least out of the three data centers.

3.3 Cost Estimations

The other objective of this study was to look at the cost of each type of data center, in comparison to how efficient they are, and to pinpoint tradeoffs. Energy costs are calculated with Equation 4:

$$Cost = \exists \times Rate \qquad (4)$$

With the total energy consumption values from Equation 2 and the Georgia Power Commercial Rate, estimated at $0.115 per kilowatt-hour [2], the total cost of each data center can be totaled. This, however, produced unexpected results: the AI system's overall energy cost was lower than both the Modern and Legacy systems. This contradicts the widely held assumption that AI data centers are expensive to build and maintain.

3.3.1Power Usage Effectiveness

Researching more into popular metrics for measuring the energy usage of data centers underlined the ideas of Power Usage Effectiveness (PUE) and IT productivity Per Embedded Watt (IT-PEW). Both offer more complexity than the measure originally used in this study, with PUE being the more widely adopted of the two. PUE is a representation of the ratio between the total power consumed by a data center and the power used by the computer servers [16]. Statista records annual average PUE numbers, shown in Figure 3, so this measure was ultimately chosen.

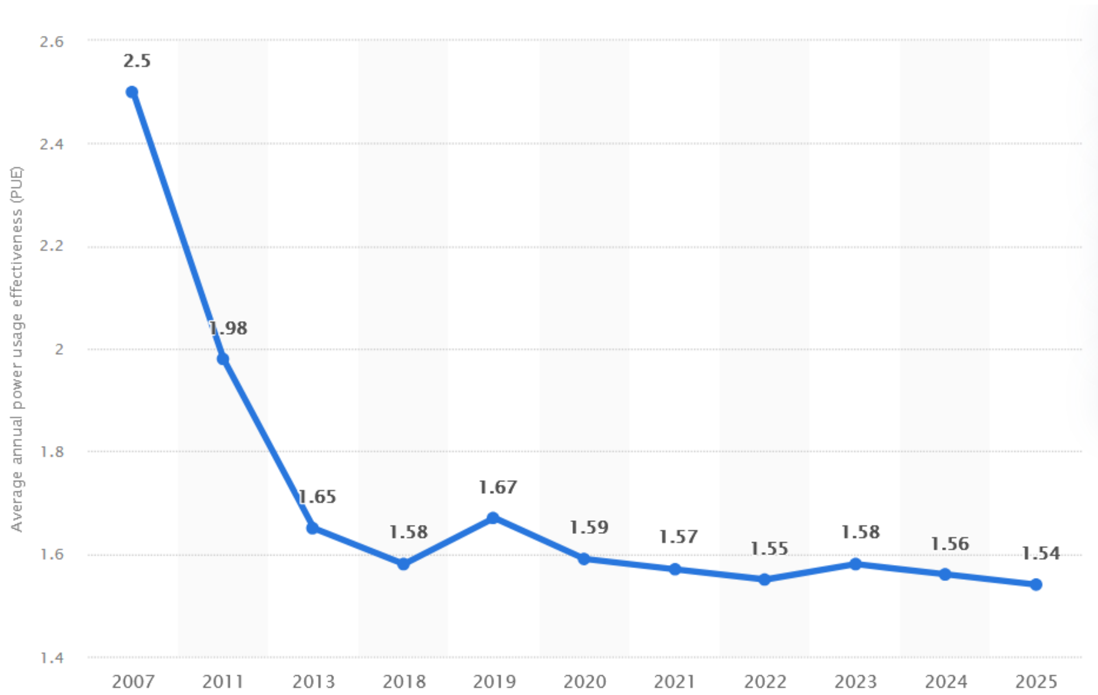

Figure 3: Data center average annual power usage effectiveness (PUE) worldwide from 2007 to 2025 [14].

Table 4 shows the PUE numbers for each data center system, sourced and utilized as close to the benchmark years for each.

Table 4: Simulation Data Center PUE values

Legacy (2000)	2.5 PUE (2007)
Modern (2017)	1.58 PUE (2018)
AI (2024)	1.56 PUE (2024)

By adding PUE to our original simulation energy consumption formula, a new layer of complexity is added that will more closely resemble a real-world application.

3.3.2Hardware Costs

Another factor that greatly contributes to data center costs which were omitted during the first calculations is hardware. Each data center simulated has 100 hosts/servers, so an estimated cost per system is added to the operational costs to show a different picture that makes the simulation more realistic. The server base cost was considered along with other elements needed for it to run–CPUs, Graphics Processing Units (GPU), memory, and storage. Table 5 displays the results.

Table 5: Hardware Costs per Data Center

Data Center	Hardware Cost
Legacy	≈ $56,000
Modern	≈ $650,000
AI	≈ $15,000,000

Software costs are another factor impacting the comprehensive cost of each system, but for the objective of this study, only the operational and hardware costs are considered as their resulting total meets expectations.

4 EXPERIMENTAL RESULTS

Using the CloudSim Plus framework, I was able to model three different types of data centers: legacy, modern, and AI. The data center system parameters were modeled after existing servers spanning from 2000 to 2024, and the specifications detailed in their respective benchmark tests. The framework allows for such customization of data center hosts, the main reason it was chosen for this study. After successfully executing the simulation, I obtained energy and execution time data for each data center type and then began on my main objective, observing trade-offs between energy consumption and efficiency in cloud computing data centers.

The initial analysis of each system's operational costs was surprising, considering the costly reputation AI has, as well as high energy draws data centers in the United States are reported to have. However, it provides context to why companies would consider jumping on the AI trend. By isolating the operational cost as illustrated in Figure 4, it is easy to see the advantages of using an AI data center compared to other modern options. The AI system used a fraction of the energy to complete the same number of tasks as the others. This supports a strong case for upgrading to AI-class infrastructure, given its exceptional computational efficiency.

However, operational cost alone is not a realistic measure of the true cost of an AI data center. In actuality, a lot of the cost behind this type of data center is found in the hardware's price tag. Figure 5 shows the total cost after factoring in hardware cost.

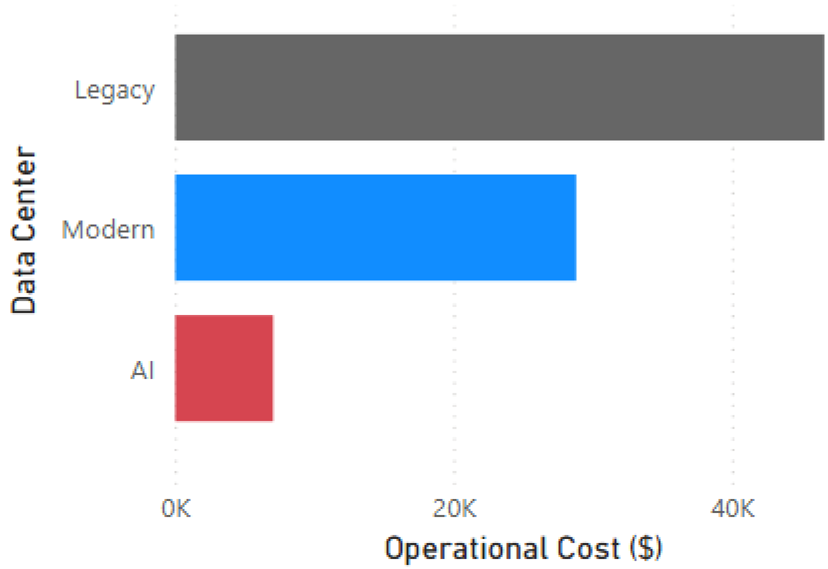

Figure 4: Cost of energy consumed by Data Center

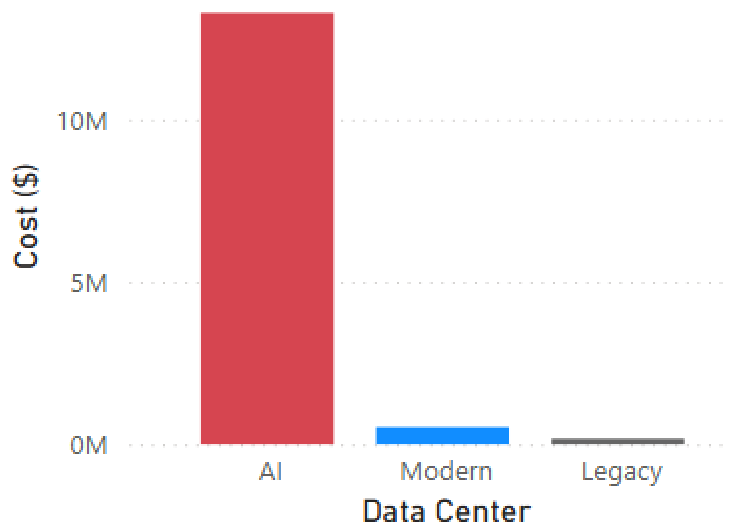

Figure 5: Overall total cost by Data Center

The throughput values of each system, displayed by Figure 6, also confirm the high efficiency of AI, as the AI system's throughput was the largest at 100,000 tasks completed per second of total time the simulation took to execute. The AI system also had the best performance per Watt numbers, indicated in Figure 7. This means it also completed the most tasks compared to the amount of energy used, represented by its low operational cost.

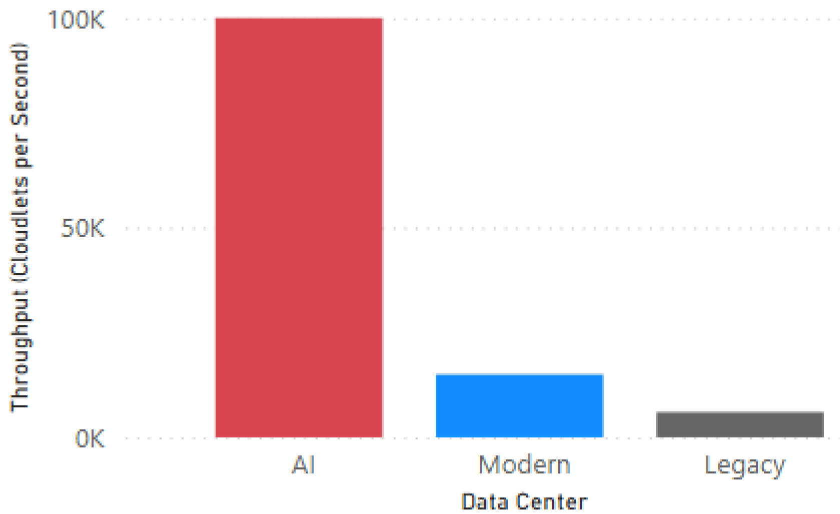

Figure 6: Throughput comparison

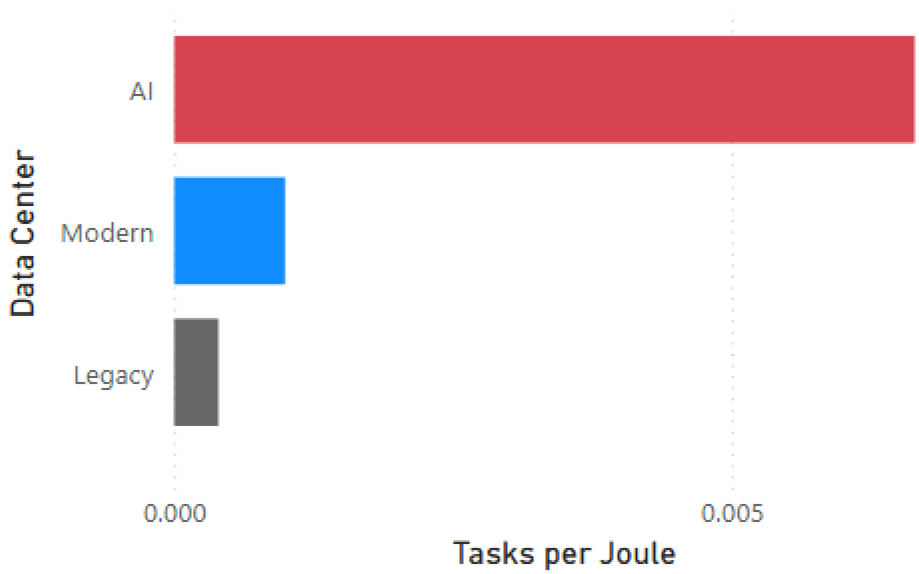

Figure 7: Performance per Watt by Data Center

5 CONCLUSION

While the AI-optimized system in the simulation demonstrates superior energy efficiency due to reduced execution time, its significantly higher hardware cost offsets these gains. When incorporating both operational and hardware costs, modern data center architecture represents the most practical middle ground. The experimental results also highlight that raw energy consumption alone is a misleading metric. Efficiency should be evaluated in the context of throughput and total cost of hardware. As AI infrastructure continues to develop, and *if* hardware costs decline, the long-term economic case for AI-optimized data centers looks optimistic.

Future work could incorporate software and environmental costs in addition to the operational and hardware costs examined here. Another cost factor that was considered but left for future work is server cost amortization, which is the spreading of upfront hardware investment over the expected lifespan of each system. Incorporating amortized costs would offer a more granular view of the long-term financial picture for each data center type. However, since satisfactory and expected results were already obtained using the power and hardware cost factors described in this study, amortization was deferred to keep the scope manageable. The `schedulers` class could also be used to add the idle power element of the original total energy used by each data center formula. AI data centers should use more power, even when idle, therefore adding that would give an even more realistic simulation.

Additionally, the simulation code itself could be modified in future work to implement dynamic task redistribution, where systems that finish their initial task list first would receive additional ones until all tasks across all data centers are complete. This could be achieved using the `addOnCloudletFinishListener()` method provided by the `DatacenterBroker` class, which provides functionality for new tasks to be dynamically submitted. Such an approach would likely result in the modern and legacy systems consuming more total power than observed in the current simulation setup, since they would remain active longer to absorb redistributed work. This was not implemented here because the current simulation already produced satisfactory results, but it represents a meaningful direction for improving simulation realism.

6 DATA AVAILABILITY

The simulation source code and dataset generated during the study are available in the following repository: https://github.com/mcarbajal2/AdvInfoSys-SemesterProject.

REFERENCES

[1] Barroso, L. A., & Hölzle, U. (2007). The case for energy-proportional computing. IEEE Computer, 40(12), 33-37.

[2] ElectricChoice.com. (n.d.). *Georgia electricity rates*. Retrieved April 21, 2026, from https://www.electricchoice.com/electricity-prices-by-state/georgia/

[3] Green, A., Nangia, I., & Sandri, N. (2025, September 9). *The infrastructure moment*. McKinsey & Company. https://www.mckinsey.com/industries/infrastructure/our-insights/the-infrastructure-moment

[4] International Energy Agency. (2025). *Energy and AI: Energy demand from AI*. https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai

[5] Koomey, J. G., Belady, C., Patterson, M., Santos, A., & Lange, K. D. (2009). Assessing trends over time in performance, costs, and energy use for servers. Lawrence Berkeley National Laboratory, Stanford University, Microsoft Corporation, and Intel Corporation, Tech. Rep.

[6] MacNeil, T. (2004, November). *Don't be misled by MIPS*. IBM Systems Magazine. https://web.archive.org/web/20120723080942/http://www.ibmsystemsmag.com/mainframe/tipstechniques/systemsmanagement/Don-t-Be-Misled-By-MIPS/

[7] Rodrigues, J., & Silva Filho, M. C. (2026). Integrated green cloud framework using virtualization, carbon-aware workload migration, and simulation for sustainable data centers. *Journal of Sustainable Computing, 14*(2), 112–128.

[8] Shehabi, A., Smith, S., Sartor, D., Brown, R., Herrlin, M., Koomey, J., ... & Lintner, W. (2016). United states data center energy usage report.

[9] Singh, A. K., Patni, S., & Saxena, D. (2025). *Resource management in cloud computing: Concepts and implementation*. Springer.

[10] Silva Filho, M. C., Oliveira, R. L., Monteiro, C. C., Inácio, P. R., & Freire, M. M. (2017, May). CloudSim plus: a cloud computing simulation framework pursuing software engineering principles for improved modularity, extensibility and correctness. In 2017 IFIP/IEEE symposium on integrated network and service management (IM) (pp. 400-406). IEEE.

[11] Standard Performance Evaluation Corporation. (2017). *SPECpower_ssj2008 result: HPE ProLiant DL360 Gen10*. https://www.spec.org/power_ssj2008/results/res2017q3/power_ssj2008-20170704-00768.html

[12] Standard Performance Evaluation Corporation. (2024). *SPECpower_ssj2008 result: HPE ProLiant DL380a Gen11*. https://www.spec.org/power_ssj2008/results/res2024q2/power_ssj2008-20240401-01391.html

[13] Stansbury, M., Marchese, K., Hardin, K., & Amon, C. (2025, June 24). *Can U.S. infrastructure keep up with the AI economy?* Deloitte Insights.

https://www.deloitte.com/us/en/insights/industry/power-and-utilities/data-center-infrastructure-artificial-intelligence.html

[14] Statista. (n.d.). *Data center average annual power usage effectiveness (PUE) worldwide*. Retrieved April 21, 2026, from https://www.statista.com/statistics/1229367/data-center-average-annual-pue-worldwide

[15] Urone, P. P., & Hinrichs, R. (2012). *College physics by OpenStax volume I & II*. XanEdu Publishing Inc.

[16] Yuventi, J., & Mehdizadeh, R. (2013). A critical analysis of Power Usage Effectiveness and its use in communicating data center energy consumption. *Energy and Buildings, 64*, 90–94.